EXPERIMENT NO. 1 A (Group A)

Aim: To use PCA Algorithm for dimensionality reduction. You have a dataset that includes measurements for different variables on wine (alcohol, ash, magnesium, and so on). Apply PCA algorithm & transform this data so that most variations in the measurements of the variables are captured by a small number of principal components so that it is easier to distinguish between red and white wine by inspecting these principal components

Name: Aditya Rade Roll.No:BEAD23167 Outputs:

In [1]:
```
%pip install scikit-learn pandas matplotlib
```

```
Requirement already satisfied: scikit-learn in c:\users\adity\appdata\local\programs
\python\python311\lib\site-packages (1.9.1)
Requirement already satisfied: pandas in c:\users\adity\appdata\local\programs\pytho
n\python311\lib\site-packages (2.2.2)
Requirement already satisfied: matplotlib in c:\users\adity\appdata\local\programs\p
ython\python311\lib\site-packages (3.11.0)
Requirement already satisfied: numpy>=1.24.1 in c:\users\adity\appdata\local\program
s\python\python311\lib\site-packages (from scikit-learn) (1.26.4)
Requirement already satisfied: scipy>=1.10.0 in c:\users\adity\appdata\local\program
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Requirement already satisfied: joblib>=1.4.0 in c:\users\adity\appdata\local\program
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Requirement already satisfied: narwhals>=2.0.1 in c:\users\adity\appdata\local\progr
ams\python\python311\lib\site-packages (from scikit-learn) (2.22.1)
Requirement already satisfied: threadpoolctl>=3.5.0 in c:\users\adity\appdata\local
\programs\python\python311\lib\site-packages (from scikit-learn) (3.7.0)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\adity\appdata\loca
l\programs\python\python311\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\adity\appdata\local\programs
\python\python311\lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: tzdata>=2022.7 in c:\users\adity\appdata\local\progra
ms\python\python311\lib\site-packages (from pandas) (2026.2)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\adity\appdata\local\prog
rams\python\python311\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\adity\appdata\local\programs
\python\python311\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\adity\appdata\local\pro
grams\python\python311\lib\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\adity\appdata\local\pro
grams\python\python311\lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in c:\users\adity\appdata\local\progr
ams\python\python311\lib\site-packages (from matplotlib) (24.2)
Requirement already satisfied: pillow>=9 in c:\users\adity\appdata\local\programs\py
thon\python311\lib\site-packages (from matplotlib) (10.4.0)
Requirement already satisfied: pyparsing>=3 in c:\users\adity\appdata\local\programs
\python\python311\lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in c:\users\adity\appdata\local\programs\pyt
hon\python311\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Note: you may need to restart the kernel to use updated packages.
```

```
[notice] A new release of pip is available: 24.0 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [2]:
```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

In [7]:
```python
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA

# Load dataset
url = "https://media.geeksforgeeks.org/wp-content/uploads/Wine.csv"
dataset = pd.read_csv(url)

# Display dataset
print("First five records:")
print(dataset.head())

print("\nDataset shape:")
print(dataset.shape)

# Separate features and target
X = dataset.iloc[:, :-1].values
y = dataset.iloc[:, -1].values

# Standardize the features
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Apply PCA
pca = PCA(n_components=2)
X_pca = pca.fit_transform(X_scaled)

# Display transformed data
print("\nPCA Transformed Data:")
print(X_pca[:5])

# Explained variance
print("\nExplained Variance Ratio:")
print(pca.explained_variance_ratio_)

print("\nTotal Variance Captured:")
print(sum(pca.explained_variance_ratio_))

# Plot PCA
plt.figure(figsize=(8, 6))
```

```
for target in set(y):
    plt.scatter(
        X_pca[y == target, 0],
        X_pca[y == target, 1],
        label=f"Class {target}"
    )

plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("PCA - Wine Dataset")
plt.legend()
plt.grid()
plt.show()
print("Name:Aditya Rade Roll No:BEAD23167")
```

```
First five records:
   Alcohol  Malic_Acid   Ash  Ash_Alcanity  Magnesium  Total_Phenols  \
0    14.23        1.71  2.43          15.6        127           2.80
1    13.20        1.78  2.14          11.2        100           2.65
2    13.16        2.36  2.67          18.6        101           2.80
3    14.37        1.95  2.50          16.8        113           3.85
4    13.24        2.59  2.87          21.0        118           2.80

   Flavanoids  Nonflavanoid_Phenols  Proanthocyanins  Color_Intensity   Hue  \
0        3.06                  0.28             2.29             5.64  1.04
1        2.76                  0.26             1.28             4.38  1.05
2        3.24                  0.30             2.81             5.68  1.03
3        3.49                  0.24             2.18             7.80  0.86
4        2.69                  0.39             1.82             4.32  1.04

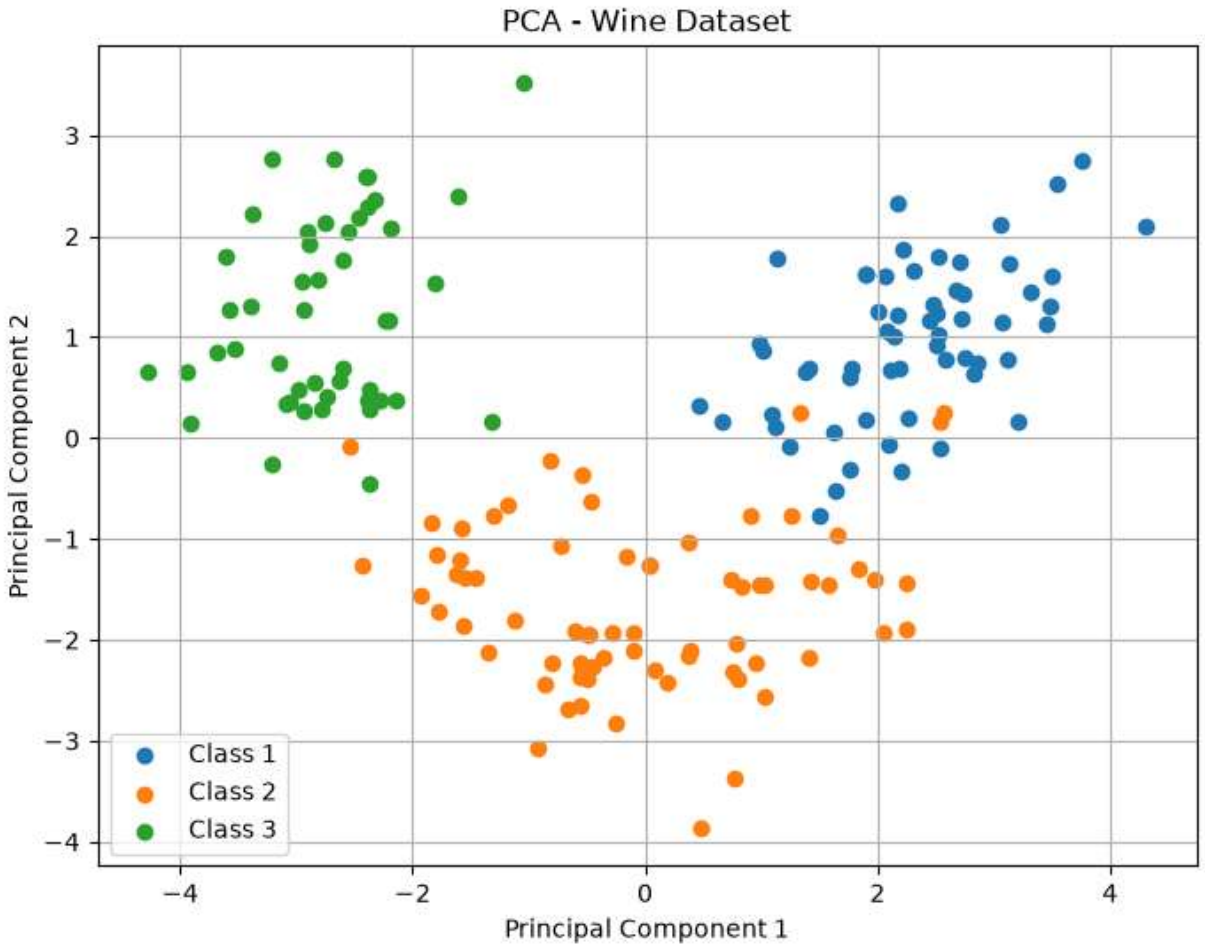
   OD280  Proline  Customer_Segment
0   3.92     1065                 1
1   3.40     1050                 1
2   3.17     1185                 1
3   3.45     1480                 1
4   2.93      735                 1

Dataset shape:
(178, 14)

PCA Transformed Data:
[[ 3.31675081  1.44346263]
 [ 2.20946492 -0.33339289]
 [ 2.51674015  1.0311513 ]
 [ 3.75706561  2.75637191]
 [ 1.00890849  0.86983082]]

Explained Variance Ratio:
[0.36198848 0.1920749 ]

Total Variance Captured:
0.5540633835693527
```



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